

# **Factory Method Design Pattern**

Lecture notes



# Factory Method: motivation

- In an application, a class can be instantiated many times in many places
  - Many opportunities for coding errors
  - Many places to make changes if the constructor method has to change
- It would be helpful to...
  - Remove knowledge of object creation from users of the object
  - Centralise (i.e. in one place only) object creation for a class

# Factory Method: in practice

- Write a class that has methods for creating objects
  - Can use a parameter to tell the method which kind of object to create
  - Overloading these methods allows for variations in the creation process
- The Command design pattern can make use of the Factory Method pattern

# Activity



## Coding

- Review the example code and watch out for:
  - how the starting point project has object creation scattered throughout the code
  - how the project with patterns uses a factory method to locate all command creation in one place
- If you have any questions, ask

Projects: `LectureExample_StartingPoint`  
and: `LectureExample_WithPatterns`